MASTER 2 · SESSION 3 / 7

Causal Inference

Session 3 — Difference-in-Differences

Alassane Diallo

Albert School · Master 2

Academic year 2026–2027

ALBERT SCHOOL

Where we left randomized experiments

Big Idea 1 — When we cannot randomize, look at time

In the previous chapter, randomization created the counterfactual comparison **by design**: a lottery decided who received tutoring, so hidden motivation could not decide it.

But there was one assumption behind everything:

We were able to control treatment assignment.

Now suppose a school introduces tutoring **without a lottery**. We cannot create comparability any more. But we have something new:

data from before and after the intervention.

The tutoring example, without a lottery

Big Idea 1 — When we cannot randomize, look at time

In 2026, **School A** introduces tutoring for all its students. **School B** does not. Nobody drew lots: School A's management decided.

We observe each school's average exam score before (2025) and after (2026):

	2025 before	2026 after
School A (tutoring)	60	70
School B (no tutoring)	68	72

Our question is the one of the whole course, asked about School A:

What would School A have scored in 2026 without tutoring?

That number, $Y_{A,2026}(0)$, is the missing counterfactual. How can we estimate it?

Naive comparison 1: across schools, after tutoring

Big Idea 2 — Two naive comparisons, two different biases

Compare the two schools in 2026:

$$70 - 72 = -2.$$

Did tutoring *lower* scores by 2 points?

No: School A was already **8 points behind** in 2025, before any tutoring. This comparison mixes the effect with **pre-existing differences** between the schools: different students, different neighbourhoods, perhaps different motivation.

Across schools

removes: changes over time
keeps: **differences between schools**

This is the selection problem of the first chapter: the treated school is not a random school.

Naive comparison 2: before and after, in School A

Big Idea 2 — Two naive comparisons, two different biases

Compare School A with itself:

$$70 - 60 = +10.$$

Did tutoring raise scores by 10 points?

Not necessarily: between 2025 and 2026 **many things changed** for everybody: the exam may have been easier, a new curriculum, a national programme. School B, without tutoring, also gained 4 points.

Before and after

removes: differences between schools
keeps: **changes over time**

Each naive comparison removes one problem and keeps the other. So the natural question is:

Can we combine the two comparisons so that both problems cancel?

Borrow School B's change, not School B's level

Big Idea 3 — Use the control group's change as the counterfactual change

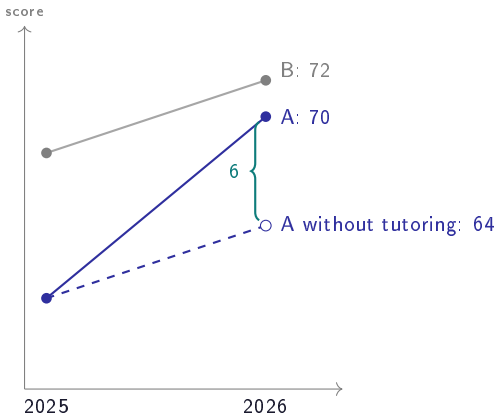
School B tells us how scores **change** without tutoring: $+4$.

If School A, without tutoring, would have changed like School B:

$$\widehat{Y}_{A,2026}(0) = 60 + 4 = 64.$$

School A actually scored 70, so

$$\text{effect} = 70 - 64 = 6 \text{ points}$$



The dashed line is the counterfactual path: School A's level, moved by School B's change.

The difference in differences

Big Idea 3 — Use the control group's change as the counterfactual change

Written as one formula, the estimate is a **difference of two differences**:

$$\widehat{\mathrm{DiD}} = (\bar{Y}_{A,\text{after}} - \bar{Y}_{A,\text{before}}) - (\bar{Y}_{B,\text{after}} - \bar{Y}_{B,\text{before}}) = (70 - 60) - (72 - 68) = 6$$

The same number can be read the other way round, as the change in the gap between the schools:

$$\underbrace{(70 - 72)}_{\text{gap after}} - \underbrace{(60 - 68)}_{\text{gap before}} = -2 - (-8) = 6.$$

	before	after	change
School A	60	70	+10
School B	68	72	+4
gap A−B	−8	−2	+6

But what exactly does this 6 measure, and what did the two subtractions remove?

Two subtractions, two problems removed

Big Idea 4 — What DiD removes, and what it estimates

First subtraction (over time)

removes everything about School A that is **stable**: its students' background, its neighbourhood, a constant level of motivation. The gap of -8 disappears.

Second subtraction (across schools)

removes everything that **changed for both schools**: an easier exam, a national reform. The common $+4$ disappears.

What remains is what changed **only in School A, only after 2025**: tutoring, if nothing else happened there.

DiD removes stable differences between groups and shocks common to both groups.

The estimand: the effect on the treated

Big Idea 4 — What DiD removes, and what it estimates

With potential outcomes $Y_{g,t}(1)$ and $Y_{g,t}(0)$ for school g in year t, the target is

$$\text{ATT} = E\big[Y_{A,\text{after}}(1) - Y_{A,\text{after}}(0)\big]$$

the effect of tutoring **on School A, in 2026**: an ATT, as in the first chapter.

- $Y_{A,\text{after}}(1) = 70$ is observed (consistency: A was tutored).
- $Y_{A,\text{after}}(0)$ is the missing counterfactual. DiD replaces it by $Y_{A,\text{before}}$ + School B's change.

DiD does not tell us what tutoring would do in School B, or in every school: it answers a question about **the treated**.

So when is the replacement correct? Let us look at what the formula really computes.

Recall: selection bias, before and after

Big Idea 4 — What DiD removes, and what it estimates

MATHEMATICS

Last chapter: observed gap = effect on the treated + selection bias. Apply it to School A versus School B **in each year** (b = before, a = after):

After (2026), A is tutored and B is not:

$$E[Y_{A,\mathrm{a}}] - E[Y_{B,\mathrm{a}}] = \underbrace{E[Y_{A,\mathrm{a}}(1) - Y_{A,\mathrm{a}}(0)]}_{\text{ATT}} + \underbrace{E[Y_{A,\mathrm{a}}(0)] - E[Y_{B,\mathrm{a}}(0)]}_{\text{selection bias after, } SB_{\mathrm{a}}}.$$

Before (2025), nobody is tutored, so the gap is *pure* selection bias, and it is **observed**:

$$E[Y_{A,\mathrm{b}}] - E[Y_{B,\mathrm{b}}] = \underbrace{E[Y_{A,\mathrm{b}}(0)] - E[Y_{B,\mathrm{b}}(0)]}_{\text{selection bias before, } SB_{\mathrm{b}}} = 60 - 68 = -8.$$

What it means. The lottery made selection bias *zero*. DiD does not: it only needs selection bias to be the **same** before and after.

In the tutoring example. School A's students start 8 points behind. If that handicap is stable, it cancels.

$$\text{DiD} = \text{after gap} - \text{before gap} = \text{ATT} + (SB_{\mathrm{a}} - SB_{\mathrm{b}})$$

Proof: what the DiD formula really measures

Big Idea 4 — What DiD removes, and what it estimates MATHEMATICS

Write $\Delta_g(d) = E[Y_{g,\text{after}}(d) - Y_{g,\text{before}}(0)]$. Before 2026 nobody is tutored, so every observed "before" score is a $Y(0)$ (consistency and no anticipation). After, School A shows $Y(1)$ and School B shows $Y(0)$.

$$\begin{aligned}\text{DiD} &= E\big[Y_{A,\text{after}}(1) - Y_{A,\text{before}}(0)\big] - E\big[Y_{B,\text{after}}(0) - Y_{B,\text{before}}(0)\big] \\ &= \underbrace{E\big[Y_{A,\text{after}}(1) - Y_{A,\text{after}}(0)\big]}_{\text{ATT}} + \underbrace{E\big[Y_{A,\text{after}}(0) - Y_{A,\text{before}}(0)\big] - E\big[Y_{B,\text{after}}(0) - Y_{B,\text{before}}(0)\big]}_{\text{difference in untreated trends}}\end{aligned}$$

(add and subtract $E[Y_{A,\text{after}}(0)]$ in the first line).

$$\text{DiD} = \text{ATT} + (\text{A's trend without tutoring} - \text{B's trend})$$

Numbers. If School A, without tutoring, would have gained $+7$ (say, catching up after a bad year), then $\text{ATT} = 70 - 67 = 3$ and the bias is $7 - 4 = 3$: the formula still says 6.

Parallel trends

Big Idea 5 — The key assumption: parallel trends

The proof shows exactly what we need: the second term must be zero.

$$E\left[Y_{A,\text{after}}(0) - Y_{A,\text{before}}(0)\right] = E\left[Y_{B,\text{after}}(0) - Y_{B,\text{before}}(0)\right]$$

Without tutoring, School A's average score would have changed by the same amount as School B's.

What it does *not* require: the two schools may have different **levels** (60 versus 68). DiD tolerates selection on levels.

What it does require: the same **change** without tutoring. It concerns $Y(0)$ after 2025 in School A, which is never observed: it cannot be tested directly.

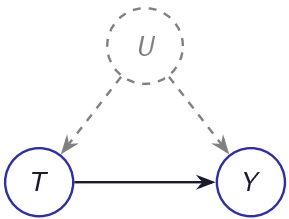
A new identifying assumption for a hidden confounder

Big Idea 5 — The key assumption: parallel trends

Remember hidden motivation U. With adjustment, a hidden U broke identification.

With DiD, School A may have **lower average motivation** than School B, and nobody needs to measure it, provided its effect on scores is the **same in 2025 and in 2026**. Then it is part of the stable gap, and the first subtraction removes it.

What DiD cannot handle is a hidden factor that **changes** differently in the two schools.



DiD replaces "no hidden confounding" by "no hidden difference in trends".

Proof: identification under parallel trends

Big Idea 5 — The key assumption: parallel trends

MATHEMATICS

Assumptions. (1) Consistency. (2) No anticipation: before 2026, scores are $Y(0)$ in both schools. (3) Parallel trends. **Claim:** the population DiD equals the ATT.

Proof. By (1) and (2), the observed means are potential means (b = before, a = after):

$$E[Y_{A,\mathrm{b}}] = E[Y_{A,\mathrm{b}}(0)], \quad E[Y_{A,\mathrm{a}}] = E[Y_{A,\mathrm{a}}(1)],$$
$$E[Y_{B,t}] = E[Y_{B,t}(0)], \ t = \mathrm{b}, \mathrm{a}.$$

The decomposition gives DiD = ATT + (difference in untreated trends); (3) sets the last term to zero:

$$E[Y_{A,\mathrm{a}} - Y_{A,\mathrm{b}}] - E[Y_{B,\mathrm{a}} - Y_{B,\mathrm{b}}] = \mathrm{ATT} \ \blacksquare$$

Every term on the left is observed: the ATT is **identified**.

Equivalently, the **change** in selection bias vanishes. Rearranging (3):

$$E[Y_{A,\mathrm{a}}(0)] - E[Y_{B,\mathrm{a}}(0)] = E[Y_{A,\mathrm{b}}(0)] - E[Y_{B,\mathrm{b}}(0)],$$

i.e. $SB_{\mathrm{a}} - SB_{\mathrm{b}} = 0$. The observed 2025 gap (-8) stands in for the 2026 selection bias: $-2 = \mathrm{ATT} + (-8)$, so $\mathrm{ATT} = 6$.

If School A would have caught up anyway, its handicap shrinks: $SB_{\mathrm{a}} \neq SB_{\mathrm{b}}$.

Proof: a model in which parallel trends holds

Big Idea 5 — The key assumption: parallel trends MATHEMATICS

Suppose untreated scores are the sum of a school part, a year part and noise:

$$Y_{it}(0) = \alpha_i + \lambda_t + \varepsilon_{it}, \qquad E[\varepsilon_{it} \mid \text{school}] = 0.$$

α_i may depend on anything stable about student i (ability, background, **motivation**), and may be correlated with tutoring. Then for each school g:

$$E\big[Y_{g,\mathrm{a}}(0) - Y_{g,\mathrm{b}}(0)\big] = E[\alpha_i - \alpha_i \mid g] + (\lambda_\mathrm{a} - \lambda_\mathrm{b}) = \lambda_\mathrm{a} - \lambda_\mathrm{b},$$

the same for A and B: parallel trends holds, **whatever the correlation between α_i and tutoring**.

When it fails. If the hidden factor's effect changes over time, e.g. $Y_{it}(0) = \alpha_i + \lambda_t + \gamma_t U_i + \varepsilon_{it}$ with $\gamma_\mathrm{a} \neq \gamma_\mathrm{b}$, the trend becomes

$$\lambda_\mathrm{a} - \lambda_\mathrm{b} + (\gamma_\mathrm{a} - \gamma_\mathrm{b})\, E[U \mid g],$$

which differs between schools when $E[U \mid A] \neq E[U \mid B]$.

Proof: DiD with covariates (conditional parallel trends)

Big Idea 5 — The key assumption: parallel trends MATHEMATICS

If School A and School B differ in composition (more scholarship students in A), trends may be parallel only **for similar students**. Write $D = 1$ for School A, $\Delta Y = Y_a - Y_b$, X pre-tutoring covariates, $e(X) = P(D = 1 \mid X) < 1$, $p = P(D = 1)$.

Assumption: $E[\Delta Y(0) \mid D = 1, X] = E[\Delta Y(0) \mid D = 0, X]$. Since $\text{ATT} = E[\Delta Y \mid D{=}1] - E[\Delta Y(0) \mid D{=}1]$, we need the second term.

Regression form. Iterate expectations over X among School A's students, then use the assumption:

$$E[\Delta Y(0) \mid D{=}1] = E\big\{E[\Delta Y(0) \mid D{=}1, X] \mid D{=}1\big\} = E\big\{E[\Delta Y \mid D{=}0, X] \mid D{=}1\big\}.$$

Predict the untreated change from School B's students with the same X, then average over School A's students:

$$\text{ATT} = E[\Delta Y \mid D{=}1] - E\big\{E[\Delta Y \mid D{=}0, X] \mid D{=}1\big\}$$

Proof: DiD with covariates, the weighting form

Big Idea 5 — The key assumption: parallel trends MATHEMATICS

Same assumption and notation. Instead of predicting the untreated change, reweight School B's students so that they look like School A's.

Weighting form (Abadie, 2005). Because $f(x \mid D{=}1) = e(x)f(x)/p$,

$$E\left[\tfrac{(1-D)\,e(X)}{1-e(X)}\,\Delta Y\right] = E\big[e(X)\,E[\Delta Y \mid D{=}0, X]\big] = p\,E\big\{E[\Delta Y \mid D{=}0, X] \mid D{=}1\big\},$$

and $E[D\,\Delta Y] = p\,E[\Delta Y \mid D{=}1]$. Subtracting and dividing by p:

$$\text{ATT} = \frac{1}{p}\,E\Big[\frac{D - e(X)}{1 - e(X)}\,\Delta Y\Big]$$

School A's students get weight 1; a School B student gets $-e(X)/(1-e(X))$, large when that student "looks like" School A. The first equality uses Bayes' rule, $f(x \mid D{=}1) = e(x)f(x)/p$, and needs $e(X) < 1$ (overlap).

Proof: parallel trends depends on the scale

Big Idea 5 — The key assumption: parallel trends MATHEMATICS

Parallel trends is an assumption about **one particular scale** of the outcome. If it holds in levels, it generally fails in logs (percentage changes), and conversely.

Our numbers. In levels, without tutoring both schools gain 4 points: $60 \to 64$ and $68 \to 72$. In logs:

$$\ln \tfrac{64}{60} = 0.065 \qquad \neq \qquad \ln \tfrac{72}{68} = 0.057.$$

General statement. Let $m_{g,t} = E[Y_{g,t}(0)]$ and h a strictly increasing transformation. Parallel trends in levels says $m_{A,\mathrm{a}} - m_{A,\mathrm{b}} = m_{B,\mathrm{a}} - m_{B,\mathrm{b}} = c$. For $h = \ln$ (and means of $h(Y)$ approximated by h of means):

$$\ln m_{A,\mathrm{a}} - \ln m_{A,\mathrm{b}} = \ln\left(1 + \frac{c}{m_{A,\mathrm{b}}}\right) \neq \ln\left(1 + \frac{c}{m_{B,\mathrm{b}}}\right) \quad \text{whenever } m_{A,\mathrm{b}} \neq m_{B,\mathrm{b}},\ c \neq 0.$$

Consequence. If instead trends are parallel in percentages, School A's counterfactual is $60 \times \frac{72}{68} = 63.5$, and the effect is 6.5, not 6.

Choose the scale on substantive grounds before estimating: levels and logs answer different questions.

Can we check an assumption we cannot test?

Big Idea 5 — The key assumption: parallel trends

Parallel trends is about a world we never observe. But it makes a prediction about a world we **do** observe: the years *before* tutoring.

If the two schools were already moving in parallel before 2026, the assumption becomes more credible.

How can past data support, or undermine, parallel trends?

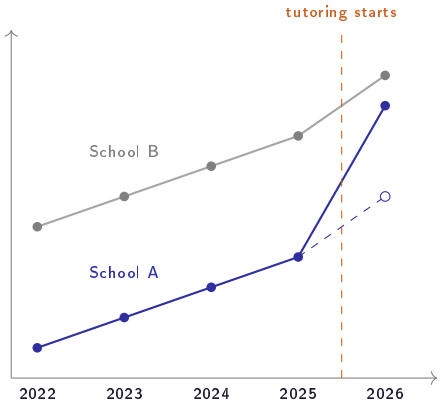
Pre-trends

Big Idea 6 — Look at the past: pre-trends and event studies

Suppose we have the scores of both schools from 2022 to 2026.

Before tutoring, both schools gain 2 points a year and the gap stays at -8. In 2026 the gap jumps to -2.

The past supports parallel trends: without tutoring, School A would plausibly have stayed 8 points behind.



A **placebo DiD** between two pre-tutoring years must be close to zero: $(60 - 58) - (68 - 66) = 0$.

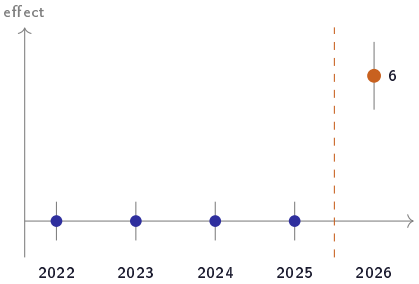
The event study

Big Idea 6 — Look at the past: pre-trends and event studies

Plot the gap A−B year by year, relative to the last year before tutoring (2025):

year	22	23	24	25	26
gap	−8	−8	−8	−8	−2
vs 2025	0	0	0	**0**	+6

Flat before, a jump after: the picture we hope to see.



Caution. Parallel pre-trends *support* the assumption; they do not *prove* it. Something could still happen in School A exactly in 2026. And a dip just before the programme (tutoring introduced *because* scores fell) is a warning: scores might have rebounded anyway.

Proof: what pre-trend checks test

Big Idea 6 — Look at the past: pre-trends and event studies MATHEMATICS

Placebo DiD. Take two years $s < t$ *both before* tutoring. Then both schools are untreated in both years, so all four means are $Y(0)$ and

$$\mathrm{DiD}_{s\to t} = E\big[Y_{A,t}(0) - Y_{A,s}(0)\big] - E\big[Y_{B,t}(0) - Y_{B,s}(0)\big].$$

If the untreated trends are parallel at all dates, this is exactly zero. A placebo DiD far from zero is evidence **against** parallel trends.

Event-study regression. With $A_i = 1$ for School A and the last pre-year $t^* = 2025$ as reference:

$$Y_{it} = \alpha_i + \lambda_t + \textstyle\sum_{k\neq t^*} \beta_k\, A_i\, 1\{t = k\} + \varepsilon_{it}$$

Each β_k is a DiD between year k and 2025. For $k < t^*$, β_k is a placebo (expected 0 under parallel trends); for $k \geq 2026$, β_k is the effect in year k.

Limit. The test concerns the past. Parallel trends after 2025 is an assumption about the future of a counterfactual world.

Proof: each event-study coefficient is a 2×2 DiD

Big Idea 6 — Look at the past: pre-trends and event studies MATHEMATICS

Two schools, years $t = 1, \ldots, K$, reference year t^* (2025). The event-study model

$$Y_{it} = c + a\,A_i + \lambda_t + \sum_{k \neq t^*} \beta_k\, A_i 1\{t = k\} + \varepsilon_{it} \qquad (\lambda_{t^*} = 0)$$

has $1 + 1 + (K-1) + (K-1) = 2K$ parameters for $2K$ cells (school × year): it is **saturated**, so OLS reproduces every cell mean:

$$\bar{Y}_{B,t} = c + \lambda_t, \qquad \bar{Y}_{A,t} = c + a + \lambda_t + \beta_t \quad (\beta_{t^*} = 0).$$

Subtract within each year, then compare with the reference year:

$$\bar{Y}_{A,k} - \bar{Y}_{B,k} = a + \beta_k, \qquad \bar{Y}_{A,t^*} - \bar{Y}_{B,t^*} = a$$

$$\hat{\beta}_k = \left(\bar{Y}_{A,k} - \bar{Y}_{B,k}\right) - \left(\bar{Y}_{A,t^*} - \bar{Y}_{B,t^*}\right)$$

Check. $\hat{\beta}_{2026} = (70 - 72) - (60 - 68) = 6$; $\hat{\beta}_{2024} = (58 - 66) - (60 - 68) = 0$; $\hat{\beta}_{2022} = (54 - 62) - (60 - 68) = 0$. Each pre-period coefficient is a placebo DiD; each post-period coefficient is the effect in that year.

From four numbers to real data

Big Idea 6 — Look at the past: pre-trends and event studies

So far we worked with four averages. Real data contain thousands of students, many schools and many years, and every estimate comes with uncertainty, exactly as in the previous chapter.

How do we compute DiD on real data, and how precise is it?

The DiD regression

Big Idea 7 — DiD as a regression, and its uncertainty

For student i in year t, let $A_i = 1$ if the student is in School A, and $\text{Post}_t = 1$ in 2026. Fit

$$Y_{it} = \beta_0 + \beta_1 A_i + \beta_2 \,\text{Post}_t + \delta\,(A_i \times \text{Post}_t) + \varepsilon_{it}$$

β_1: the stable gap between schools (-8).
β_2: the common change over time ($+4$).
δ: what changed only in A, only after: the **DiD** (6).

With many schools and years, the general form is the **two-way fixed effects** regression

$$Y_{it} = \alpha_i + \lambda_t + \delta\, T_{it} + \varepsilon_{it},$$

with a unit effect α_i, a year effect λ_t, and $T_{it} = 1$ when unit i is treated in year t.

Proof: the interaction coefficient is the DiD

Big Idea 7 — DiD as a regression, and its uncertainty MATHEMATICS

With two groups and two periods, the regression has four parameters for four cells, so it fits the four cell means exactly (OLS reproduces cell means in a saturated model):

	before (Post $= 0$)	after (Post $= 1$)
School B ($A = 0$)	$\beta_0 = 68$	$\beta_0 + \beta_2 = 72$
School A ($A = 1$)	$\beta_0 + \beta_1 = 60$	$\beta_0 + \beta_1 + \beta_2 + \delta = 70$

Subtract down each column and then across:

$$\big[(\beta_0 + \beta_1 + \beta_2 + \delta) - (\beta_0 + \beta_1)\big] - \big[(\beta_0 + \beta_2) - \beta_0\big] = (\beta_2 + \delta) - \beta_2 = \delta.$$

$$\hat{\delta} = (\bar{Y}_{A,\mathrm{a}} - \bar{Y}_{A,\mathrm{b}}) - (\bar{Y}_{B,\mathrm{a}} - \bar{Y}_{B,\mathrm{b}}) = (70 - 60) - (72 - 68) = 6$$

Solving: $\beta_0 = 68$, $\beta_1 = -8$, $\beta_2 = 4$, $\delta = 6$. The regression is the same arithmetic, plus standard errors.

Proof: fixed effects, first differences and DiD coincide

Big Idea 7 — DiD as a regression, and its uncertainty MATHEMATICS

Panel of students i observed before and after. Two-way fixed effects model:

$$Y_{it} = \alpha_i + \lambda_t + \delta\, T_{it} + \varepsilon_{it}, \qquad T_{it} = D_i\, 1\{t = \mathrm{a}\}.$$

Step 1: difference out the student. Subtract the before equation from the after equation:

$$\Delta Y_i = Y_{i\mathrm{a}} - Y_{i\mathrm{b}} = \underbrace{(\lambda_{\mathrm{a}} - \lambda_{\mathrm{b}})}_{\text{common change}} + \delta\, D_i + \Delta\varepsilon_i.$$

Every stable student characteristic in α_i (ability, background, motivation) has disappeared.

Step 2: regress ΔY_i on D_i. With a single binary regressor, the OLS slope is the difference of group means:

$$\hat{\delta} = \overline{\Delta Y}_{D=1} - \overline{\Delta Y}_{D=0} = (70 - 60) - (72 - 68) = 6$$

Step 3: unbiasedness. $E[\hat{\delta}] = \delta + E[\Delta\varepsilon \mid D{=}1] - E[\Delta\varepsilon \mid D{=}0]$, so $\hat{\delta}$ is unbiased exactly when the *changes* in the error do not depend on treatment: parallel trends, written in the model.

Uncertainty: shocks are shared within a school

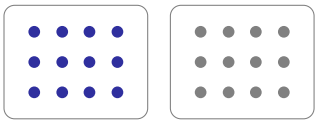
Big Idea 7 — DiD as a regression, and its uncertainty

Students of the same school in the same year share a teacher, a principal, a flu wave, a difficult exam room. Their scores move **together**.

This is the clustering problem of the previous chapter. Standard errors must be computed at the level where treatment varies: the **school** (clustered standard errors).

And with only **two schools**, adding students does not help much: we need **many treated and many control schools**.

one shared shock per school and year



Proof: why more students do not fix two schools

Big Idea 7 — DiD as a regression, and its uncertainty MATHEMATICS

Score of student i in school g, year t: $Y_{igt} = \mu_{gt} + \nu_{gt} + \varepsilon_{igt}$, where ν_{gt} is a shock shared by the whole school that year (variance σ_ν^2) and ε_{igt} is individual (variance σ_ε^2). Shocks are independent across schools and years; each cell has n students.

Each cell mean contains its school shock: $\bar{Y}_{gt} = \mu_{gt} + \nu_{gt} + \bar{\varepsilon}_{gt}$, with $\mathrm{Var}(\bar{\varepsilon}_{gt}) = \sigma_\varepsilon^2/n$. The DiD combines four cells with signs ± 1, so

$$\mathrm{Var}\left(\widehat{\mathrm{DiD}}\right) = 4\,\sigma_\nu^2 + \frac{4\,\sigma_\varepsilon^2}{n} \xrightarrow[n\to\infty]{} 4\,\sigma_\nu^2 > 0$$

More students shrink only the second term. With G treated and G control schools, the school shocks are averaged too:

$$\mathrm{Var}\left(\widehat{\mathrm{DiD}}\right) = \frac{4\sigma_\nu^2}{G} + \frac{4\sigma_\varepsilon^2}{Gn} \xrightarrow[G\to\infty]{} 0.$$

Same lesson as the design effect $1 + (m-1)\rho$: **the number of independent units that matters is the number of schools.**

Proof: following the same students sharpens DiD

Big Idea 7 — DiD as a regression, and its uncertainty MATHEMATICS

Before and after scores have variance σ^2 and, for the same student, correlation ρ (strong students stay strong). Two designs, n students per school and year:

Repeated cross-sections (new students each year): the before and after means are independent,

$$\mathrm{Var}\left(\bar{Y}_{\mathrm{a}} - \bar{Y}_{\mathrm{b}}\right) = \frac{2\sigma^2}{n}.$$

Panel (same students): $\mathrm{Cov}(\bar{Y}_{\mathrm{a}}, \bar{Y}_{\mathrm{b}}) = \rho\sigma^2/n$, so

$$\mathrm{Var}\left(\overline{\Delta Y}\right) = \frac{2\sigma^2(1-\rho)}{n}.$$

The DiD is the difference of two such changes (independent schools), so its individual-noise part is

$$\mathrm{Var}_{\mathrm{panel}} = (1-\rho) \times \mathrm{Var}_{\mathrm{cross\text{-}section}}$$

Numbers. $\sigma = 12$, $n = 1000$, $\rho = 0.8$: variance of each school's change 0.288 with new students, 0.058 with the same students: **five times** more precise (CUPED again). School-year shocks ν_{gt} are not removed: only more schools reduce them.

So far, the clean case

Big Idea 7 — DiD as a regression, and its uncertainty

So far: one programme, one start date, parallel trends, nothing else happening in School A. Real policies are messier: schools adopt at different dates, students move, other changes happen at the same time.

What can go wrong with DiD in practice?

Threats to a DiD design

Big Idea 8 — Parallel trends can be correct, while the setting goes wrong

School A introduced tutoring *because* its scores were falling; they might have rebounded anyway	→ **selection on trends**
Another change happens in School A in 2026 (a new principal)	→ **concurrent shock**
Strong students move into School A to get tutoring	→ **composition change**
Students start working harder in 2025, when tutoring is announced	→ **anticipation**
School B students attend School A's sessions or share notes	→ **spillovers**
Schools start tutoring in different years, with effects that grow over time	→ **staggered adoption**

Each one either breaks parallel trends, or makes "before" or "untreated" mean something else than we assumed.

Staggered adoption: a hidden trap in the regression

Big Idea 8 — Parallel trends can be correct, while the setting goes wrong

Suppose many schools start tutoring in different years. The two-way fixed effects regression then makes many 2×2 comparisons at once, and some of them use **already-treated schools as controls**.

If the effect of tutoring grows over time, the already-treated school's change includes **its own growing effect**, and subtracting it removes part of the effect we want, sometimes even reversing its sign.

Modern estimators (Callaway and Sant'Anna; Sun and Abraham; de Chaisemartin and D'Haultfœuille) compare each newly treated group only with **not-yet-treated or never-treated** groups.

DiD solves stable confounding, not every problem of an observational design.

Proof: the forbidden comparison, in numbers

Big Idea 8 — Parallel trends can be correct, while the setting goes wrong MATHEMATICS

All schools share a common trend of $+3$ points a year. **School A** starts tutoring in 2025: its effect is 4 in 2025 and grows to 8 in 2026. **School C** starts in 2026, with effect 6.

Compare C (newly treated) with A (already treated) between 2025 and 2026:

$$\text{change in C} = \underbrace{3}_{\text{trend}} + \underbrace{6}_{\text{C's effect}} = 9,$$

$$\text{change in A} = \underbrace{3}_{\text{trend}} + \underbrace{(8-4)}_{\text{growth of A's effect}} = 7,$$

$$\text{DiD} = 9 - 7 = 2 \neq 6.$$

In general, when the control group is already treated,

$$\text{DiD} = \text{effect on the newly treated} - \text{change in the controls' own effect}$$

The bias vanishes only if the effect is constant over time. Comparing C with a never-treated school (change 3) gives $9 - 3 = 6$: correct.

Theorem: what two-way fixed effects averages

Big Idea 8 — Parallel trends can be correct, while the setting goes wrong MATHEMATICS

With staggered adoption, groups k start treatment at different dates; U denotes never-treated groups.

Theorem (Goodman-Bacon, 2021). With a balanced panel, the TWFE coefficient is a weighted average of all 2×2 DiDs between pairs of groups:

$$\hat{\delta}_{\text{TWFE}} = \sum_k s_{kU}\,\hat{\delta}^{2\times2}_{kU} + \sum_{k\neq l} s_{kl}\,\hat{\delta}^{2\times2}_{kl}, \qquad s \geq 0,\ \sum s = 1$$

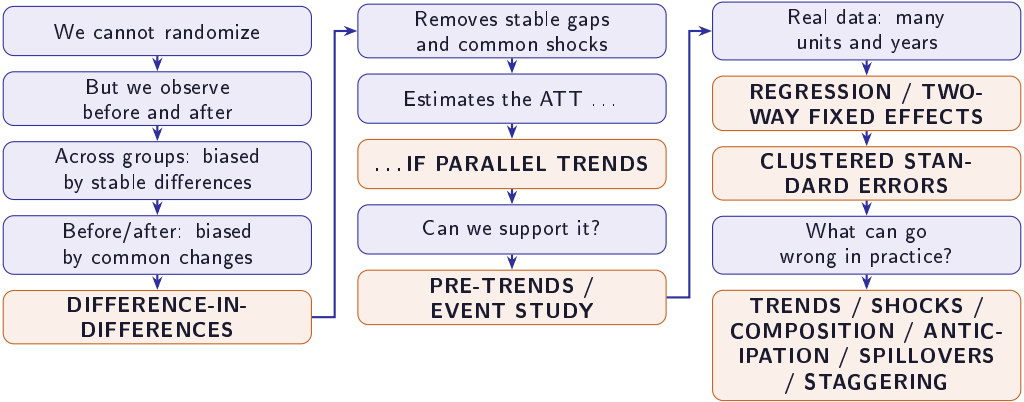
The weights grow with group size and with the **variance of treatment** in each pair's window, $\bar{D}(1-\bar{D})$.

Why it matters. Some of the $\hat{\delta}^{2\times2}_{kl}$ compare a *later*-treated group with an *earlier*-treated one used as control. The previous slide showed such a comparison equals *effect on the newly treated − change in the controls' own effect*: biased when effects grow over time (2 instead of 6). In terms of the true group-time effects, some implicit weights become **negative** (de Chaisemartin and D'Haultfœuille, 2020).

Remedy. Estimate each group-time effect against never- or not-yet-treated groups, then average with positive weights (Callaway and Sant'Anna, 2021).

The complete logic of difference-in-differences

Big Idea 9 — The complete logic of difference-in-differences



The big message

Big Idea 9 — The complete logic of difference-in-differences

Difference-in-differences builds the counterfactual from the control group's change over time.

Randomization made the groups comparable **in levels**, by design.

DiD accepts groups that differ in levels, and asks them to be comparable **in changes**.

The price is a new assumption, parallel trends, which no data can fully verify.

But what if trends are not parallel?

Big Idea 10 — But what if trends are not parallel?

Everything in this chapter relied on one assumption:

Without treatment, the treated and control groups would have evolved in parallel.

Sometimes this is not credible: the hidden factor that pushed School A into tutoring also changes its trend.

Remember noncompliance in randomized experiments: the lottery **offer** changed who attended tutoring, but the offer itself could not affect scores except through attendance.

That suggests a new kind of evidence:

something that shifts treatment, but affects the outcome only through treatment.

Next chapter — Instrumental Variables

Big Idea 10 — But what if trends are not parallel?

So our next question becomes:

Can we find something that pushes some students into tutoring for reasons unrelated to their exam scores, and use it to recover the effect of tutoring?

This question introduces the next chapter:

Instrumental Variables (IV)

For now, we stop there. The next chapter will explain **how and under what assumptions** such a push can be used for causal inference.

References: Card, D. and Krueger, A. B. (1994). Minimum wages and employment. *American Economic Review* 84(4). Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics*, ch. 5. Goodman-Bacon, A. (2021) and Callaway, B. and Sant'Anna, P. (2021), *Journal of Econometrics* 225(2). Roth, J., Sant'Anna, P., Bilinski, A. and Poe, J. (2023). What's trending in difference-in-differences? *Journal of Econometrics* 235(2).